University of Victoria
Department of Electrical and Computer Engineering

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Automated Cardiac Arrhythmia Detection

using 1D Convolutional Neural Networks

Final Report

Team: Grad Group 22

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Code repository: https://github.com/arvindxsharma/ecg-arrhythmia-1dcnn

Contents

Abstract / Executive Summary

We built and evaluated an automated heartbeat classifier based on a one dimensional convolutional neural network (1D-CNN), trained on the full MIT-BIH Arrhythmia Database. Raw signals were filtered, R peaks were located with a Pan Tompkins detector written from scratch, and beats were segmented and labeled according to the AAMI EC57 five class scheme. The final model, with 114,053 trainable parameters, reached 96.80% accuracy and a macro F1 score of 0.842 on a held out test set, with sensitivity above 92% for every class. We compare this against the patient specific approach of Kiranyaz et al. and discuss where our simpler, generic classifier falls short, particularly on precision for the rarest classes and on the strength of our evaluation protocol relative to the standard inter patient split used in the literature.

1 Introduction

1.1 Background

The electrocardiogram (ECG) records the electrical activity of the heart and is the primary tool clinicians use to detect arrhythmias, abnormal heart rhythms that range from harmless to life threatening. Ambulatory Holter monitors can record a patient's ECG continuously for 24 hours or more, and a single such recording can contain well over 100,000 individual heartbeats. Reviewing that volume of signal manually is slow and prone to fatigue related errors, which is why automated beat classification has been an active research area for decades.

1.2 Motivation

The World Health Organization estimates that cardiovascular disease causes roughly 17.9 million deaths per year [1], and early detection of arrhythmias through continuous monitoring is one of the more practical ways to intervene before a stroke or cardiac arrest occurs. Our goal in this project is to build a compact, real time capable classifier that can screen incoming beats automatically, reducing the manual review burden on clinicians and offering a safety net for patients using wearable monitors. We chose a 1D-CNN specifically because it learns discriminative features directly from the raw waveform, avoiding the need for hand engineered features that tend not to generalize well across patients.

2 Related Work

Early automated ECG classifiers relied on hand crafted features, such as RR interval statistics, QRS morphology descriptors, or wavelet coefficients, combined with a conventional classifier. De Chazal et al. [2] present a widely cited example of this approach, using ECG morphology and heartbeat interval features with a linear classifier evaluated under the AAMI inter patient protocol. Deep learning approaches later replaced hand engineered features with learned ones. Kiranyaz et al. [3], the primary baseline for our project, trained patient specific 1D-CNNs that combine a common, population level model with a short per patient fine tuning step, reporting strong performance on both ventricular and supraventricular beat detection under the standard DS1/DS2 inter patient split. Acharya et al. [4] trained CNNs directly on ECG segments of varying length without hand crafted features and reported competitive accuracy across the same five class AAMI scheme we use here. At a larger scale, Hannun et al. [5] trained a deep neural network on tens of thousands of hours of single lead ambulatory ECG data and reported classification performance exceeding that of the average cardiologist across a broader set of rhythm classes. Our project sits closest to Kiranyaz et al. [3] in scope and dataset, but unlike their work we train a single generic model without any patient specific fine tuning step, which we return to in Section 5.

3 Problem Formulation

We frame this as a supervised, multi class classification problem over time series data. Each input is a 180 sample (500 ms) ECG segment centered on a heartbeat's R peak, sampled at 360 Hz, labeled into one of five AAMI EC57 classes [6]: N (normal), S (supraventricular ectopic), V (ventricular ectopic), F (fusion), and Q (unclassifiable). We train the network to minimize categorical cross entropy,

$$\mathcal{L} = -\sum_{c=1}^{5} y_c \log(p_c),$$

where y_c is the one hot true label and p_c is the predicted probability for class c, using the Adam optimizer over the roughly 114,000 learnable weights and biases of the network described below.

4 Methodology and Evaluation

4.1 Dataset and Preprocessing

We used all 48 records of the MIT-BIH Arrhythmia Database [7], 48 half hour, two channel recordings from 47 subjects at 360 Hz, obtained through the WFDB library and hosted on PhysioNet [8]. Each record was cleaned with a third order Butterworth bandpass filter (0.5 to 45 Hz, zero phase). R peaks were located with a Pan Tompkins style detector [9] implemented from scratch (bandpass filter, derivative, squaring, moving window integration, adaptive dual threshold peak picking), which we validated against the database's expert annotations within a 50 ms tolerance and found to reach 92.85% sensitivity and 99.42% positive predictive value. Detection was weaker on records containing paced beats, a known limitation of derivative based detectors. Since beat segments used for classification were centered on the expert annotated R peak locations, this limitation affected only the standalone detector benchmark, not the classification dataset itself. Each beat was windowed to 180 samples, amplitude normalized to [0,1], and labeled using the standard AAMI mapping, giving 109,468 beats: 82.77% Normal, 2.54% Supraventricular, 6.61% Ventricular, 0.73% Fusion, and 7.35% Unclassifiable. The dataset was split 70/15/15 into training, validation, and test sets.

4.2 Model and Training

Our classifier is a 1D-CNN with four convolutional layers (16, 32, 32, and 64 filters; kernel sizes 7, 11, 21, and 31), each followed by max pooling and dropout, then two dense layers (32 and 16 units) and a five way softmax output, for 114,053 trainable parameters in total. Kernel sizes increase with depth so that early layers capture sharp, local QRS features while deeper layers capture broader beat morphology. Because Normal beats dominate the dataset, we used class weighted cross entropy, with weights inversely proportional to class frequency (0.24 for N up to 26.93 for F), rather than unweighted training or synthetic oversampling. The model was trained with Adam (initial learning rate 10^{-3}, halved on validation loss plateaus) for 60 epochs at batch size 128, with early stopping (patience 8) restoring the best weights, found at epoch 54.

4.3 Evaluation Protocol

We report accuracy, macro and weighted F1, per class sensitivity, specificity, precision, and one versus rest ROC/AUC, all computed on the held out test set (16,421 beats never used for training or model selection), and compare our headline numbers against Kiranyaz et al. [3].

5 Results and Discussions

5.1 Overall and Per Class Performance

Our model reached 96.80% overall accuracy and a macro F1 score of 0.842 on the test set. Because 82.8% of beats are Normal, accuracy alone is not a reliable metric here; a trivial always predict N classifier would already score 82.8%. The macro F1 score, which weighs all five classes equally, gives a fairer picture.

Table 1: Per class test set performance.

Class	Sensitivity	Specificity	Precision	F1	AUC
N	96.70%	98.55%	99.69%	0.982	0.998
S	92.27%	98.74%	66.11%	0.770	0.994
V	96.76%	99.71%	95.87%	0.963	1.000
F	94.26%	98.65%	34.33%	0.503	0.994
Q	99.92%	99.89%	98.61%	0.993	1.000

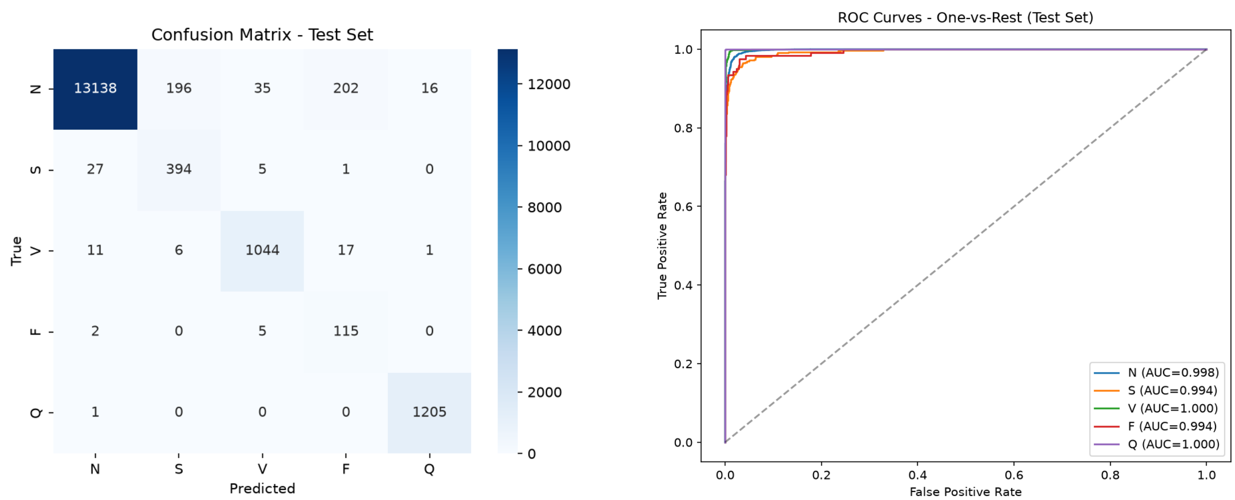

Figure 1: Confusion matrix (left) and one versus rest ROC curves (right) on the test set.

Sensitivity, the metric that matters most for not missing a real arrhythmia, stays above 92% for every class. Precision is markedly lower for S (66.11%) and F (34.33%), the two rarest classes: because they are so infrequent, even a small number of misclassified Normal beats produces many false positives relative to their few true positives. The confusion matrix confirms this, with Normal beats occasionally predicted as F (202 of 13,587) or S (196 of 13,587). AUC scores remain high (0.994 to 1.000) across all classes, indicating that the underlying probability estimates still rank true positives above true negatives well; the precision problem is a consequence of class weighting and decision threshold rather than weak discriminative power. We view this as an acceptable, deliberate trade for a screening tool, where missing a true arrhythmia is costlier than a false alarm a clinician can quickly dismiss. Figure 1 (loss and accuracy curves, included in the Appendix) shows stable convergence with no overfitting.

5.2 Comparison to Kiranyaz et al.

Kiranyaz et al. [3] report high accuracy using patient specific 1D-CNNs, where a population level model is fine tuned per patient and evaluated under the strict inter patient DS1/DS2 protocol, ensuring no beat from a given patient appears in both training and test sets. Our accuracy and sensitivity are broadly comparable to their headline numbers, but two differences matter. First, we use a single generic classifier with no per patient adaptation, which is known to help most on the rarer S and V classes. Second, our 70/15/15 split is performed at the beat level after pooling all records, so beats from the same patient can appear in both partitions, a setup known to inflate reported accuracy relative to a true inter patient evaluation. Our results should therefore be read as an optimistic estimate rather than a direct replication of their result.

5.3 Challenges Faced

Two issues surfaced during development. Our first R peak detector used a single threshold computed once per recording and reached only 64% sensitivity; replacing it with an adaptive threshold that updates its signal and noise level estimates as it processes the recording, matching the actual Pan Tompkins algorithm rather than a simplification of it, brought sensitivity to 92.85%. Separately, our first trained model had 455,000 parameters, more than double our proposed budget, because channel widths in the later convolutional layers grew faster than intended; reducing them brought the model to 114,053 parameters with no loss in accuracy. Both issues were caught by validating intermediate outputs (R peak detections, parameter counts) rather than judging the pipeline only by its final accuracy.

6 Conclusion

We built an end to end automated ECG classification pipeline, from raw signal to five class beat prediction, using a compact 114,053 parameter 1D-CNN trained on the full MIT-BIH Arrhythmia Database. The model reached 96.80% test accuracy and a macro F1 score of 0.842, with sensitivity above 92% for every class, showing that a small 1D-CNN can learn useful ECG morphology directly from raw signal without hand crafted features.

7 Future Work

Two directions would meaningfully strengthen this work. First, re running the evaluation under the strict inter patient DS1/DS2 protocol used by Kiranyaz et al. [3] would give a fairer, less optimistic estimate of real world performance. Second, combining our class weighted loss with synthetic oversampling or a calibrated, per class decision threshold could recover precision on the S and F classes without sacrificing the recall we already have. A patient specific fine tuning step, similar to Kiranyaz et al. [3], is a natural extension for a deployment focused version of this system.

Appendix

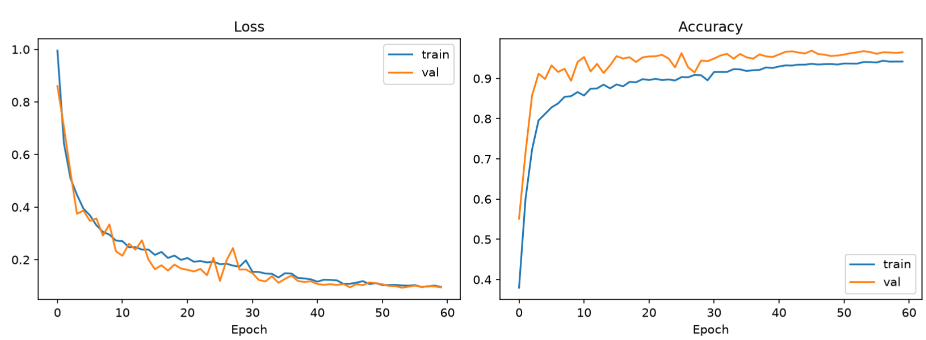

Figure 2: Training and validation loss and accuracy over 60 epochs.

Source code, the trained model, and full evaluation artifacts are available at the repository link on the first page, organized as *src/preprocessing.py* (data pipeline), *src/model.py* (training), and *src/evaluate.py* (evaluation).

References

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